

SATNA, India

WMO: 425710

Lat: 24.567N

Lon: 80.833E

Elev: 317

StdP: 97.57

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.5	8.8	-1.2	3.6	28.4	0.8	4.2	25.9	2.5	19.3	2.3	19.7	0.0	320	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.0	43.2	22.0	42.1	22.0	40.9	21.9	27.7	32.2	27.2	31.5	27.0	31.1	1.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.1	29.5	26.3	22.6	29.1	26.1	22.3	28.8	91.0	32.4	89.2	31.7	87.7	31.1	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 3.0	(b) 2.4	(c) 2.2	(e) 5.3	(f) 45.0	(g) 1.2	(h) 1.0	(i) 4.5	(j) 45.7	(k) 3.8	(l) 46.3	(m) 3.1	(n) 46.9	(o) 2.2	(p) 47.6		
			DB													
			WB													
				5.3	45.0	1.2	1.0	4.5	45.7	3.8	46.3	3.1	46.9	2.2	47.6	
				4.1	29.1	1.2	0.7	3.2	29.6	2.5	30.0	1.8	30.4	0.9	31.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	16.6	20.4	25.6	31.5	34.6	33.5	29.1	28.3	28.4	26.6	22.3	18.0		
	DBStd	6.09	2.61	2.74	3.01	2.39	2.02	3.18	2.00	1.46	1.42	1.99	2.06	2.32		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	111	67	10	0	0	0	0	0	0	0	0	1	34		
	CDD10.0	5931	204	290	483	645	764	706	591	568	551	513	368	248		
	CDD18.3	3001	13	67	226	395	506	456	332	310	301	255	118	24		
	CDH23.3	39686	150	647	2607	6014	8453	7197	3992	3478	3360	2566	1007	214		
	CDH26.7	22113	20	203	1396	3922	6009	4851	1695	1238	1311	1148	300	22		
(14) Wind	WSAvg	0.5	0.3	0.4	0.5	0.6	0.8	0.9	0.7	0.6	0.5	0.3	0.2	0.2		
(15) Precipitation	PrecAvg	1083	19	20	15	8	15	125	310	331	181	48	8	6		
	PrecMax	1694	83	105	44	49	88	363	867	556	770	234	69	74		
	PrecMin	623	5	0	1	0	0	16	133	151	6	8	0	0		
	PrecStd	265	18	28	16	12	21	100	157	135	157	55	15	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	33.1	39.0	42.8	44.6	45.0	38.6	34.9	35.2	35.7	31.9	28.8		
		MCWB	17.5	18.3	19.4	20.5	21.4	24.1	25.5	26.0	24.8	21.1	19.6	17.7		
	2%	DB	26.8	31.3	37.0	41.5	43.5	43.2	35.8	33.2	34.0	34.3	30.6	27.0		
		MCWB	16.4	18.1	18.5	20.6	21.6	23.1	26.3	26.5	25.0	21.9	19.4	16.7		
	5%	DB	25.0	29.2	35.2	40.5	42.5	41.7	34.1	32.3	33.1	33.1	29.7	25.7		
		MCWB	15.4	17.0	18.3	20.3	21.7	23.2	26.3	26.6	25.1	22.0	19.0	16.1		
	10%	DB	23.0	27.5	33.6	39.3	41.4	40.2	32.7	31.4	32.2	32.0	28.6	24.4		
		MCWB	14.5	16.6	18.3	19.9	21.8	23.8	26.5	26.5	25.2	21.7	18.3	15.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	20.9	22.5	25.9	27.4	28.5	28.1	28.0	27.6	26.5	23.7	19.6		
		MCDB	26.2	28.5	32.3	40.1	38.2	35.4	32.5	31.6	31.2	30.7	29.3	25.9		
	2%	WB	17.5	19.5	21.1	23.1	25.8	27.4	27.6	27.5	27.1	25.7	22.3	18.5		
		MCDB	23.9	27.6	30.9	37.2	36.8	33.7	31.7	30.9	30.8	29.8	27.4	24.7		
	5%	WB	16.6	18.5	20.2	22.0	24.9	26.9	27.2	27.1	26.6	25.0	20.8	17.4		
		MCDB	22.0	26.2	30.2	35.5	36.0	33.1	31.2	30.4	30.3	28.9	26.3	23.2		
	10%	WB	15.7	17.6	19.4	21.2	24.1	26.3	26.9	26.8	26.2	24.4	19.7	16.4		
		MCDB	20.8	24.7	29.3	34.8	36.0	32.6	30.8	29.9	29.8	28.2	25.6	22.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.3	14.1	14.5	13.0	9.7	5.9	5.3	6.8	11.0	13.6	13.4		
		MCDBR	15.4	15.4	16.2	15.5	14.1	12.4	8.4	6.8	8.7	12.3	13.8	14.5		
	5% WB	MCWBR	6.5	5.3	4.6	4.7	4.0	3.4	2.6	2.2	2.5	3.1	4.7	5.7		
		MCWBR	11.7	12.9	13.5	13.9	11.9	9.0	6.4	5.9	6.6	7.7	10.9	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.485	0.472	0.490	0.554	0.628	0.680	0.633	0.586	0.559	0.572	0.579	0.532		
	taud		1.909	2.001	1.979	1.854	1.737	1.683	1.795	1.932	1.969	1.849	1.746	1.822		
	Ebn at Noon		752	805	814	770	711	671	702	736	745	705	658	684		
	Edh at Noon		173	169	181	209	235	246	220	191	180	194	203	183		
(44) All-Sky Solar Radiation	RadAvg		3.96	4.99	6.13	6.89	6.92	5.56	4.38	4.36	4.80	4.95	4.26	3.84		
	RadStd		0.38	0.30	0.28	0.23	0.20	0.51	0.46	0.36	0.52	0.32	0.24	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page